

Addressing Calculation:

192.168.10.0 $\Rightarrow$ for starting

A) 31 hosts

$$2^6 - 2 = 62 \text{ hosts}$$

192.168.10.00	00 00 00
192.168.10.00	11 11 10

192.168.10.00 | 11 11 11 $\rightarrow$ broadcast

192.168.10.0 /26

192.168.10.63 /26 $\rightarrow$ broadcast

B) 10 hosts

$$2^4 - 2 = 14 \text{ hosts}$$

192.168.10.0100	0000
192.168.10.0100	1110

1111 $\rightarrow$ brood cast

192.168.10.64 /28

192.168.10.79 /28 $\rightarrow$ broadcast

C) 10 hosts

$$2^4 - 2 = 14 \text{ hosts}$$

192.168.10.0101	0000
192.168.10.0101	1110

1111

192.168.10.80 /28

192.168.10.95 /28

D) 2 hosts

$$2^2 - 2 = 2 \text{ hosts}$$

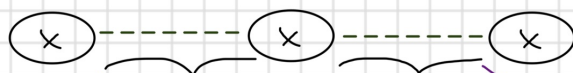
192.168.10.0110	0000
192.168.10.0110	0010

11

192.168.10.96 /30

192.168.10.99 /30

For routers:



we have 2 routers
/30 is enough

1.0.0.4 /30

1.0.0.7 /30 $\rightarrow$ broadcast